Week 1 review: build a trustworthy research starting point

Prepared 21 October 2026. Person A: Vedant. Seven workflow stages were completed across 19-21 October. Code preparation, artifact checking and initial annotation drafts were assisted by Codex; A has verified and accepted the day-seven annotations. Independent B review and joint scientific decisions remain open.

Current status: Person A's setup and reviewed development-baseline phase is complete. Joint week-one sign-off remains open where independent review or shared scientific decisions are required.

The overall goal

Goal: build a reproducible way to study whether stored KV-cache quantization changes factual disclosure from an already-unlearned language model.

The KV cache is temporary attention memory reused while generating later tokens. Here quantization compresses that stored memory while the model's weights stay fixed within a comparison.

Why it mattered: a changed answer could come from a different checkpoint, prompt, mask, numerical precision, broken cache path or scoring error. We first needed to measure and control those factors.

How we approached it: establish the GPU setup, compare full/retain/unlearned checkpoints, review factual answers, check utility, integrate upstream evaluation code, validate real quantized cache use, then repeat the baseline in a shared environment.

Final week-one output: a reviewed ordinary-cache development baseline, fixed inputs and annotations, verified cache-path evidence for the tested configuration, measured baseline cost, and a concrete next development experiment. We have not yet measured quantization-induced forgotten-fact recovery.

1. Get the model running on the intended hardware

Goal: produce one traceable answer on a Tesla T4.

Why it mattered: a connected notebook kernel alone does not establish that the intended model, tokenizer and GPU work together.

How we achieved it: used Colab, recorded environment and immutable revisions, rendered a real TOFU prompt, and saved tokens, generation settings and output.

Result: the full model correctly returned Hsiao Yun-Hwa's name. The prompt had 70 tokens; the response had 14 tokens including EOS and took 3.742 seconds. Package conflicts and restart requirements were recorded. Later notebooks corrected the distinction between base-model defaults and the effective generation configuration.

2. Establish full-model and retain-only reference behavior

Goal: understand what the full and retain90 checkpoints answer on the same 20 forgotten-author development questions.

Why it mattered: the retain-only model can answer some questions without having trained on that author. We must measure that reference behavior instead of assuming it is zero.

How we achieved it: fixed prompts, tokenizer and greedy settings; saved 40 outputs; reviewed requested facts, behavior, reference quality and prompt-supplied information.

Result: the initial full/retain correct counts were 16/20 and 3/20. Exact string matching would have reported only 13/20 and 0/20. Semantic review was therefore necessary. Ambiguous and partial cases remained explicit rather than being silently removed.

3. Add the selected unlearned engineering candidate

Goal: establish an ordinary-cache unlearned baseline before compression.

Why it mattered: later cache effects need to be compared with that same checkpoint's ordinary behavior. Checkpoints must not be selected because compression later gives a desirable result.

How we achieved it: used the previously recorded SimNPO checkpoint on the identical 20 questions and reviewed it with the same rubric.

Result: SimNPO had 2 correct, 2 partial, 15 incorrect and 1 ambiguous answer. One response reached the common 64-token budget and was retained. These observations concern one exposed author, not the entire method. SimNPO remains an engineering candidate; NPO/RMU remain the planned paper core.

4. Check whether the model still answers useful questions

Goal: screen retained/general performance rather than judging the model only from forgotten-author answers.

Why it mattered: low forgotten-fact disclosure is uninformative if the model broadly fails at ordinary tasks.

How we achieved it: froze a 60-item/model sample before generation: 30 retained questions from ten authors, 15 real-author questions and 15 world-fact questions. Reviewed 180 outputs for correctness, behavior, coherence and relevance.

Result: the selected sample did not show pervasive general incoherence. The current accepted counts remain:

Utility split	Items/model	Full correct	Retain90 correct	SimNPO correct
Retained authors	30	19	23	24
Real authors	15	12	14	11
World facts	15	11	13	11

Full and retain90 each have one additional partial retained answer. This restricted screen is not the official `model_utility` metric or a representative estimate of all retained knowledge.

5. Find and control the attention-mask problem

Goal: check that upstream evaluation helpers receive the intended tokens and scoring targets.

Why it mattered: an evaluation convention could change answers before any compression and then be mistaken for a quantization effect.

How we achieved it: ran pinned upstream preprocessing, collation, probability and ROUGE helpers on the same exposed questions. Compared the stock mask with an all-ones mask on unpadded inputs.

Result: the padding ID equals the real end-of-turn ID, so the collator masks two genuine prompt markers. Changing only the mask changed full/retain90/SimNPO text on 5/20, 14/20 and 19/20 questions. All 60 all-ones generations exactly reproduced earlier text and tokens. New integration uses explicit valid-token masks. Changed text alone is not recovered knowledge.

6. Verify that the quantized cache is actually used

Goal: confirm stored quantization, reconstructed values, attention consumption and consistent scoring/generation.

Why it mattered: unchanged scores can hide an inactive or unobserved intervention; changed scores alone do not prove the intended cache path.

How we achieved it: combined B's HQQ diagnostics with the day-six T4 run. Compared ordinary parallel/sequential scoring, an identity cache, instrumented/plain INT4 and INT2 caches, empty-cache prefill, one-token fixtures, fresh-cache repeats and native generation replay. Traced cache-returned tensors into SDPA.

Result: all 28 day-six checks passed in the tested FP32/HQQ, unpadded batch-one configuration. All 16 layers showed reconstructed K/V reaching attention. Identity, instrumentation and empty-cache parallel comparisons had zero score differences; the largest ordinary parallel/sequential gap was 0.00003052 NLL per token.

Artificial fixture	Cache	Mean sequential NLL change	Largest absolute token change
Short	INT4	-0.41685	1.10800
Short	INT2	-0.01325	2.51839
Long	INT4	-0.10546	0.99159
Long	INT2	+0.33273	3.10312

The empty-cache parallel scores were unchanged, while later sequential scores changed. This demonstrates the scoped evaluation-path mechanism on artificial prompts. It does not establish factual recovery, universal cache behavior or literature novelty. Three repeat checks archived only derived deltas; that evidential limit is recorded.

7. Build and accept the shared-environment TOFU baseline

Goal: move the real development questions into the numerical environment that passed cache validation.

Why it mattered: earlier baseline runs used Torch 2.11/FP16, whereas the validated path uses Torch 2.9.1/FP32. A quantized comparison needs a matching ordinary reference.

How we achieved it: ran the same 80 questions across full, retain90 and SimNPO, producing 240 schema-v2 records. Scored the 20 forget answers per model in parallel and sequentially, including EOS. Compared new outputs with previous text/tokens and reviewed the five changes. A explicitly accepted the full annotation review.

Result: all 240 records completed without errors; all 60 ordinary score comparisons passed. 235 outputs match earlier text and tokens. The full model changed an award answer, reducing its fully correct forget count from 16 to 15. Current A-accepted counts are:

Forget label, all 20 items retained	Full	Retain90	SimNPO
Correct	15	3	2
Partial	1	3	2
Incorrect	3	13	15
Ambiguous	1	1	1

The borderline SimNPO philosophy partial was accepted as proposed. Reference issues and unsupported elaborations remain documented. One unchanged SimNPO response is truncated. The batch took 356.29 seconds (about 5.94 minutes) and reported 4.69 GiB peak allocated memory per model, loaded sequentially. Those costs describe ordinary inference, not quantized deployment performance.

What Person B contributed and what we learned from problems

Problem / contribution	Why it mattered	Outcome
B's INT4/INT2, long/short and residual-policy diagnostics	Established real cache writes/reads and implementation behavior	HQQ ran on T4; supporting reports are preserved
Quanto extension build failed on the attempted T4 path	Backend support cannot be assumed from a dependency list	Use the tested HQQ path for the next limited integration run
FP16 ordinary-path tolerance failure	Numerical baseline error could be mistaken for cache effects	FP32 reduced the fixture discrepancy; the new baseline uses the validated environment
Source mask/EOS collision	Changes outputs without compression	Explicit valid-token masks; preserve stock comparator separately
Ambiguous/non-exhaustive references and generic targets	String overlap is insufficient for factual disclosure	Accepted semantic rubric; retain partial/ambiguous/reference flags
Original B runtime reset lost reports	Temporary runtimes are not durable archives	Committed files are explicit reruns; preserve downloads and failure provenance
Multiple setup changes between early and current runs	Output drift cannot be attributed to precision alone	Current ordinary baseline is authoritative for the next paired comparison

Week-one completion and remaining joint work

Deliverable	Current status
Person A's GPU setup, provenance, baseline runs and review	Completed for the selected development scope
A's day-seven annotation verification	Accepted; AI assistance and carry-forward origins preserved
Utility screen and baseline feasibility measurements	Completed for the fixed restricted sample
Cache/scoring integration	Passed the programmed tested-scope checks; broader coverage limits documented
Independent B review of A's labels and shared evidence	Still pending; script success is not a substitute
Final checkpoint/confirmation allocation and effect/utility/numerical policy	Joint decisions remain open
Main quantized disclosure/utility experiment and paper findings	Not yet completed

We can close Person A's first-week setup/baseline phase and start the next development task. We cannot declare every joint week-one sign-off or the research pipeline universally complete.

Measurement and data handoff

- **Primary planned comparison:** correct forgotten-fact disclosure under INT4 minus ordinary cache on a fixed question population. Report all raw label categories and both directions of answer changes.

- **References:** run full and retain90 under the same cache conditions. Report the unlearned change and retain-reference change separately; their difference is secondary and can be misleading if the reference deteriorates.

- **Scores:** preserve parallel/sequential reference log-probability sums, target counts and EOS conventions. Predictive scores and ROUGE are secondary evidence, not direct knowledge measurements.

- **Development allocation:** the 20 Hsiao Yun-Hwa questions are exposed development data. The existing utility sample is exposed too. The remaining forget authors are not assigned to confirmation by this review; audit exposure and jointly freeze that allocation first.

- **Statistics:** future uncertainty must preserve author clusters and model/condition pairing. One forgotten author cannot support a population confidence interval. Multiple checkpoints/hyperparameters are not independent training seeds.

- **Open policy:** practical effect size, acceptable utility deterioration, final tolerances, annotation independence and multiplicity handling remain to be justified jointly. This review does not invent them.

The next executable step

The next development notebook and input ZIP prepare a same-run ordinary/INT4 comparison on the existing 80 questions across the three existing checkpoints. Both cache conditions use the same loaded FP32 model, masks, token IDs, generation settings and scoring conventions. Ordinary responses are regenerated within that run, so both sides have current paired controls.

The notebook contains an initial traced probe for each model, then saves both conditions incrementally. This is limited development evidence; NPO/RMU research scope and final confirmation choices are unchanged. Return the output ZIP for artifact checking and proposed paired factual review. Positive recovery is not required. No numerical or factual result from this next notebook is claimed here because it has not yet run on the GPU.

Where to inspect the evidence

- Setup and earlier reviews: `results/day1/` through `results/day6/`; B's reports: `results/diagnostics/`.

- Current baseline: `results/day7/day7_20261021T101236202224Z/`.

- Authoritative current labels: `manual-review/day7-annotation-A-accepted.csv` and `manual-review/acceptance.json` inside that run.

- Method and open gates: `docs/BLUEPRINT.md`, `docs/VALIDATION.md`, `docs/PILOT_CRITERIA.md`.

- Practical next-stage instructions: `results/int4-development-preparation/README.md`.

The original raw files, proposed reviews and earlier status snapshots are preserved. Acceptance records establish current annotation status; they do not imply independent second review or joint scientific sign-off.